AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Design basis: examples/sic-buck-400v-200v-5kw/design/design_decisions.json (schema 3, 2026-10-10) — single source of truth; electrical model automation/kicad_tools/sic_buck_control_model.py; generator automation/kicad_tools/generate_sic_schematics.py.
400 V (360-440 V) to 200 V / 25 A / 5 kW / 50 kHz synchronous buck; Wolfspeed C4MS025120K; TMS320F28335 control board on a stacked SELV mezzanine.
Net classes: CTRL_SELV and Default 0.2 mm (35 um copper), CTRL_PWR for the supply rails (1.0 mm tracks, added assumption A-G2); the shared HV patterns are guards and match no net of this SELV board.
Board rules (.kicad_dru): the baseline class-pair rules are carried as guards; the control board must never carry HV copper and its copper never lies above HV copper of the power board (air clearance >= 11 mm, standards review item).
Polarity adaptations A-C6 (UVLO flag inverted to UVLO_N) and A-C7 (raw PG_BIAS_x_N inverted before the AND combiner) use one SN74LV04A; the power board keeps the raw PG polarity on mezzanine pins 30/32/33.
Evidence: native ERC manifests under verification/erc/<run>/run.json; review PDF under review/; this drawing has no CAD evidence beyond those runs. AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED.

12 V entry, TVS and TPS26400 eFuse

文件: 01_aux_12v.kicad_sch

5 V buck, dual LDO, supervisor and reset

文件: 02_supplies.kicad_sch

TMS320F28335 — all 176 pins

文件: 03_dsp.kicad_sch

DSP per-pin bypass and ADC reference

文件: 04_dsp_decoupling.kicad_sch

Oscillator, boot straps, JTAG, UART

文件: 05_clock_debug.kicad_sch

PWM gating, fault latch, combiners, comparators

文件: 06_pwm_fault_logic.kicad_sch

PWM gating, fault latch, combiners, comparators (2)

文件: 06_pwm_fault_logic_2.kicad_sch

ADC conditioning, NTC readout, monitors

文件: 07_adc_frontend.kicad_sch

Mezzanine header, connectors, LEDs, test points, mounting

文件: 08_interconnect.kicad_sch

AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

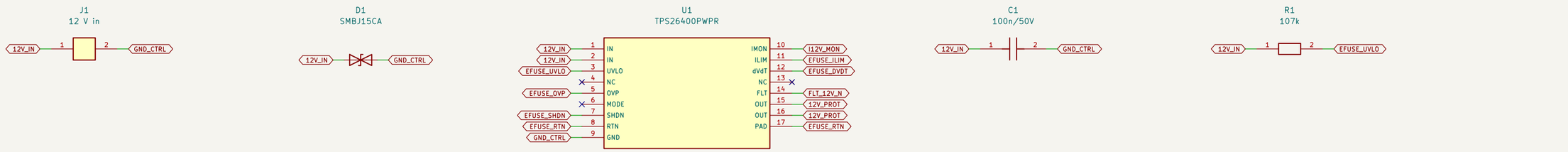
Sheet: /
File: AIPE_SiC_Buck_Control.kicad_sch

Title: AIPE V2 SiC buck control board — 400 V (360-440 V) to 200 V / 25

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 1/10

12 V entry, TVS and TPS26400 eFuse

TPS26400 eFuse: UVLO 10.0 / 9.3 V and OVP 14.45 / 13.4 V from 107 k / 4.42 k / 10.0 k, R_ILIM 6.04 k (I_CB about 1.96 A), C_dVdT 100 nF, R_IMON 14.7 k, MODE open = circuit breaker (CR−18). RTN (EFUSE_RTN) is never tied to GND_CTRL (A−C3).
PWR_FLAG symbols state supply provenance for ERC; they do not prove voltage, current capacity or protection.



AI−GENERATED ENGINEERING PROTOTYPE − V2 − HUMAN REVIEW REQUIRED

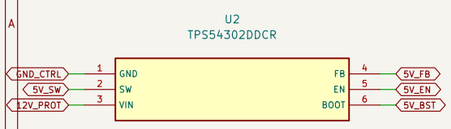
Sheet: /12 V entry, TVS and TPS26400 eFuse/
File: 01_aux_12v.kicad_sch

Title: 12 V entry, TVS and TPS26400 eFuse

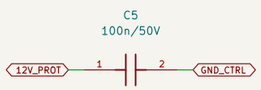
Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 2/10

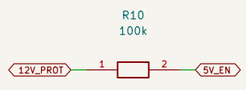
5 V buck, dual LDO, supervisor and reset

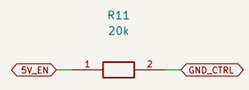
TPS54302 5 V buck (FB 100 k / 13.3 k), TPS767D301 dual LDO with core-first sequencing (1.9 V from 18.2 k / 30.1 k, 3.3 V fixed), TPS3808G01 supervisor 101 k / 10 k (4.50 V threshold, CT 10 nF). The -EP LDO variant is NRND; TLV752 is the named alternative.

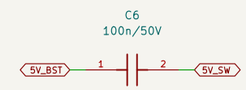










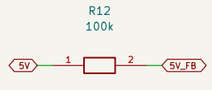






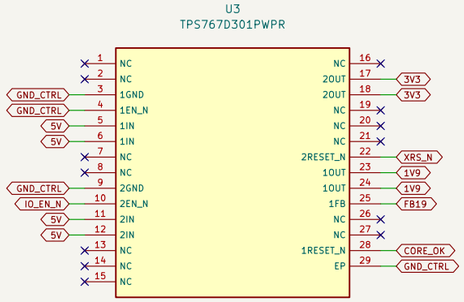








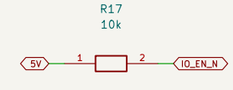






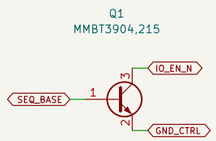


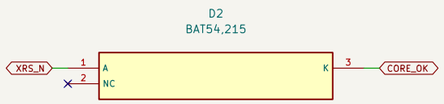




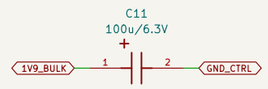




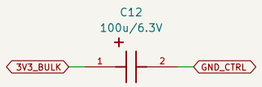






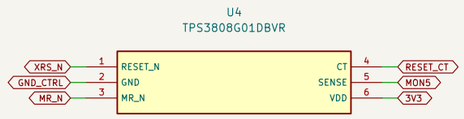










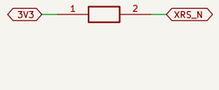


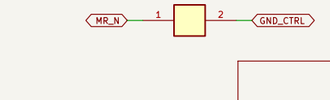












AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /5 V buck, dual LDO, supervisor and reset/
File: 02_supplies.kicad_sch

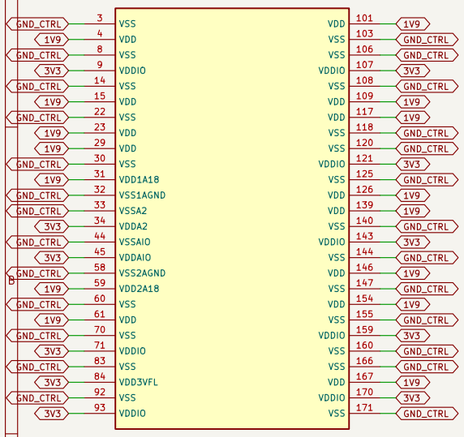
Title: 5 V buck, dual LDO, supervisor and reset

Size: A3	Date:	Rev: V2
KiCad E.D.A. 10.0.7		Id: 3/10

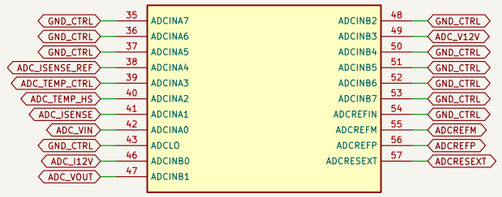
TMS320F28335 — all 176 pins

TMS320F28335PGFA, all 176 pins; units split into supplies, ADC, connected GPIO and unused pins. Pin map per design_decisions.json.

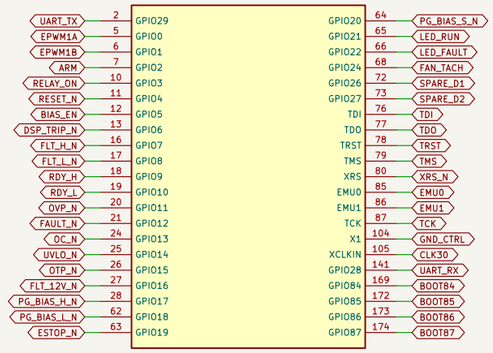
U5A
TMS320F28335PGFA



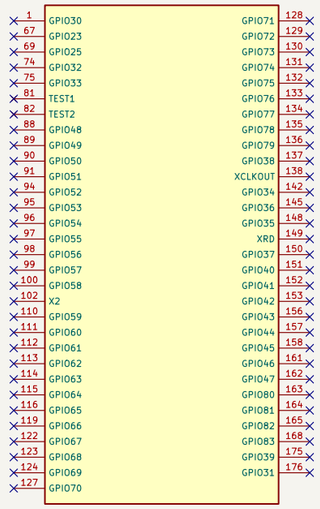
U5B
TMS320F28335PGFA



U5C
TMS320F28335PGFA



U5D
TMS320F28335PGFA



AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /TMS320F28335 — all 176 pins/
File: 03_dsp.kicad_sch

Title: TMS320F28335 — all 176 pins

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 4/10

DSP per-pin bypass and ADC reference

One 100 nF per DSP supply pin (26), 2 x 2.2 uF per rail, 2.2 uF on ADCREFP/ADCREFM, 22 k on ADCRESEXT (SPRS439Q).





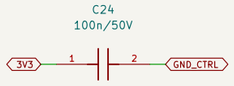






























































AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /DSP per-pin bypass and ADC reference/
File: 04_dsp_decoupling.kicad_sch

Title: DSP per-pin bypass and ADC reference

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 5/10

Oscillator, boot straps, JTAG, UART

30 MHz oscillator on XCLKIN (X1 grounded, X2 open), 33 ohm series; boot straps GPIO84–87 pulled high; TI 14-pin JTAG; SCI-A header.

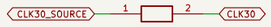
Y1
ASV-30.000MHZ-EJ-T



C49
100n/50V



R26
33



R27
4.7k



R28
4.7k



R29
4.7k



R30
4.7k



R31
4.7k



R32
4.7k



R33
4.7k



R34
4.7k



R35
2.2k



R36
10k



J3
J_JTAG: TI 14-pin, key pin 6 absent; TCK_RET tied to TCK; pin 4 TDIS to ground



J4
J_BOOT: jumper 1-2 = SCI-A boot, 2-3 or open = flash



J5
J_UART: SCI-A, 3.3 V levels only



R37
10k



AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

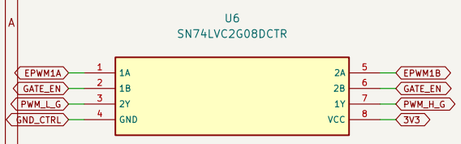
Sheet: /Oscillator, boot straps, JTAG, UART/
File: 05_clock_debug.kicad_sch

Title: Oscillator, boot straps, JTAG, UART

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 6/10

PWM gating, fault latch, combiners, comparators

ePWM1A/1B AND GATE_EN (SN74LVC2G08) -> 33 ohm -> mezzanine; SN74LVC1G74 fault latch (D = 1, CLK = ARM, CLR_N = FAULT_LATCH_N, Q = FAULT_N = ARMED); 3 x SN74LV21A combiners; SN74LV04A restores the UVLO and PG_BIAS polarities (A-C6, A-C7); TLV1702 comparators with REF3125-derived thresholds.



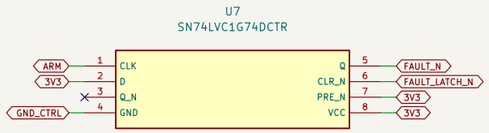




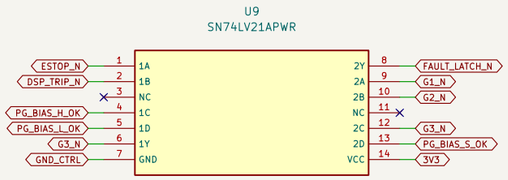




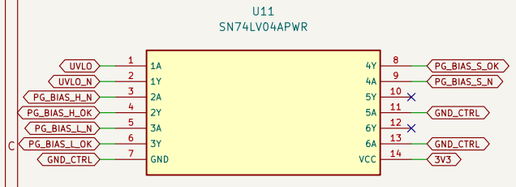




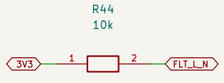


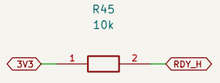
















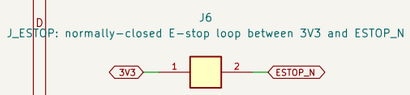


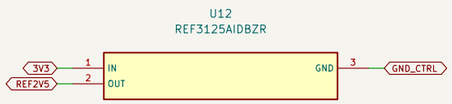
































AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /PWM gating, fault latch, combiners, comparators/
File: 06_pwm_fault_logic.kicad_sch

Title: PWM gating, fault latch, combiners, comparators

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 7/10

PWM gating, fault latch, combiners, comparators (2)

ePWM1A/1B AND GATE_EN (SN74LVC2G08) -> 33 ohm -> mezzanine; SN74LVC1G74 fault latch (D = 1, CLK = ARM, CLR_N = FAULT_LATCH_N, Q = FAULT_N = ARMED); 3 x SN74LV21A combiners; SN74LV04A restores the UVLO and PG_BIAS polarities (A-C6, A-C7); TLV1702 comparators with REF3125-derived thresholds.

R59
21.5k



C54
100n/50V



U13
TLV1702AIDGKR



U14
TLV1702AIDGKR



R60
10k



R61
10k



R62
10k



R63
10k



C55
100n/50V



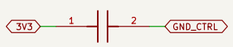
C56
100n/50V



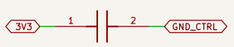
C57
100n/50V



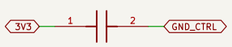
C58
100n/50V



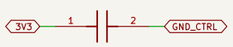
C59
100n/50V



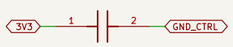
C60
100n/50V



C61
100n/50V



C62
100n/50V



AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

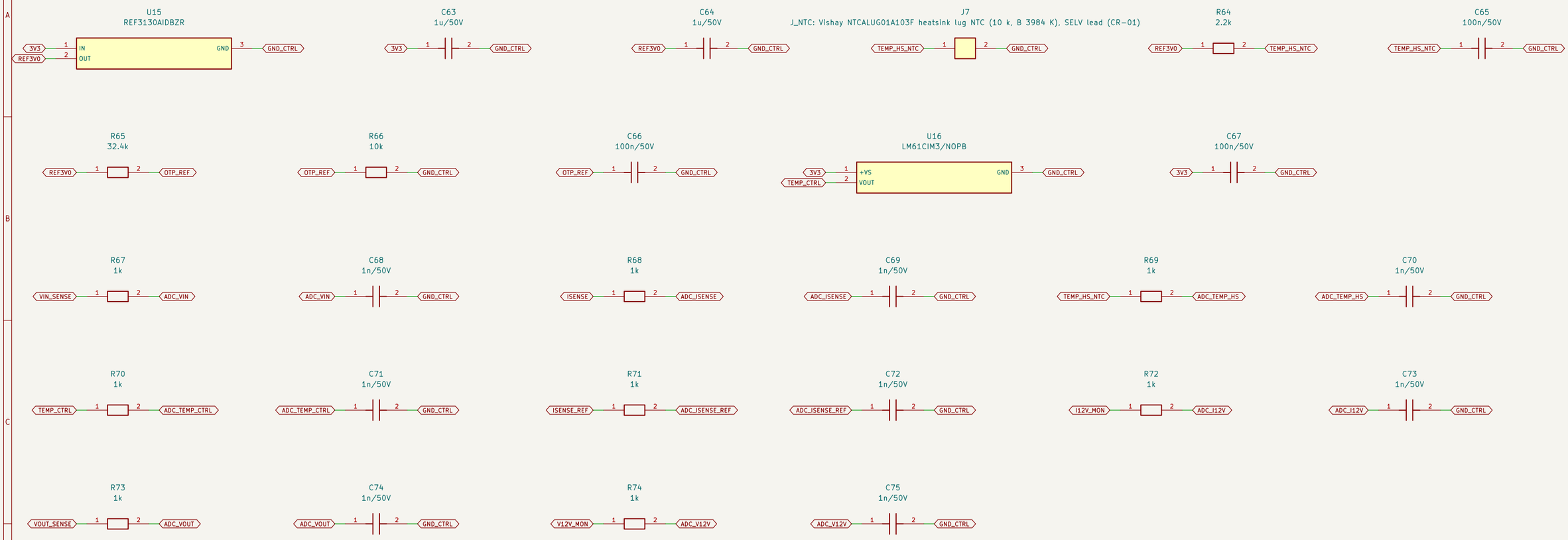
Sheet: /PWM gating, fault latch, combiners, comparators (2)/
File: 06_pwm_fault_logic_2.kicad_sch

Title: PWM gating, fault latch, combiners, comparators (2)

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 8/10

ADC conditioning, NTC readout, monitors

1 k + 1 nF at every ADC input; heatsink NTC with 2.2 k to REF3V0 (REF3130); OTP threshold 42.2 k / 10 k = 0.575 V (110 C); LM61.



AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /ADC conditioning, NTC readout, monitors/
File: 07_adc_frontend.kicad_sch

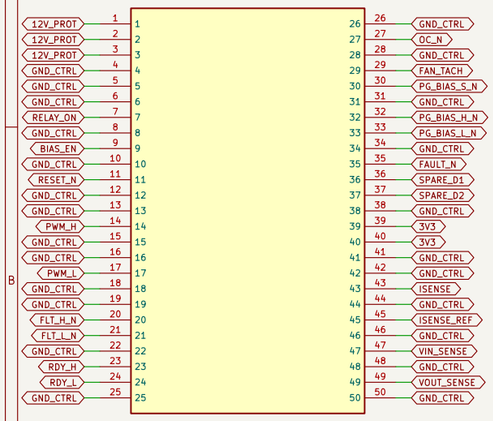
Title: ADC conditioning, NTC readout, monitors

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 9/10

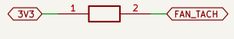
Mezzanine header, connectors, LEDs, test points, mounting

Mezzanine header TSW-125-07-G-D (pin assignment per design_decisions.json), indicator LEDs, test pads, 4 x M3.

J8
6 header to the power-board SSW-125-01-G-D socket; nets per design_decisions.json interconnect.pin_assignment



R75
10k



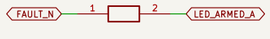
R76
4.7k



D3
LED red



R77
1.5k



D4
LED green



R78
1.5k



D5
LED green



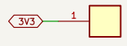
R79
1.5k



D6
LED red



TP1
3V3



TP2
1V9



TP3
5V



TP4
12V_PROT



TP5
GND_CTRL



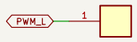
TP6
GND_CTRL



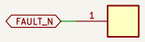
TP7
PWM_H



TP8
PWM_L



TP9
FAULT_N



TP10
ARM



TP11
FAULT_LATCH_N



TP12
GATE_EN



TP13
XRS_N



TP14
VIN_SENSE



TP15
VOUT_SENSE



TP16
ISENSE



H1
M3

H2
M3

H3
M3

H4
M3

AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /Mezzanine header, connectors, LEDs, test points, mounting/
File: 08_interconnect.kicad_sch

Title: Mezzanine header, connectors, LEDs, test points, mounting

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 10/10